GIRIJANANDA CHOWDHURY UNIVERSITY (GCU)

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DEPARTMENT OF CIVIL ENGINEERING

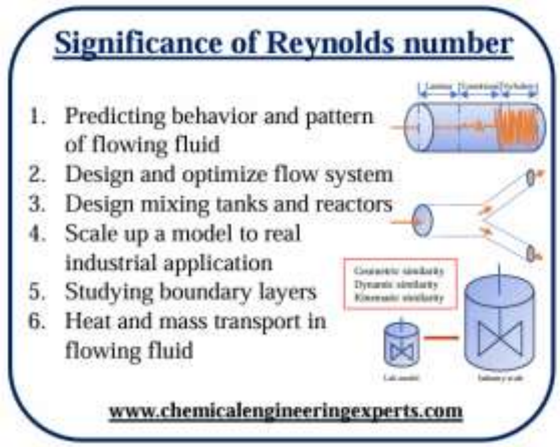
B-Tech.4th. Sem. (Regular) End Semester Examination
Practical-Viva Question Paper (Winter 2025)

Subject: Hydraulics Engineering Laboratory (BCE23414P)

Date of Examination: 14/06/2025, Saturday.

Time: 09:00 a.m.-12:00 noon **(Gr. A)**
01:00 p.m.-04:00 p.m. **(Gr. B)**

Q.1 what are the significance of Reynolds's number?



Ans: Reynolds number is very important in many aspects of engineering. It usually becomes significant when our study system involves flowing fluid. Reynolds number plays crucial role in bringing laboratory models to real life.

Predicting fluid behavior and pattern

By applying Reynolds engineers can predict flow and pattern of a flowing fluid. A calculative value of Reynolds number can give an idea about flow regime such as laminar, transition or turbulent.

Designing and optimizing flow systems

A new flow system can be designed and optimized with the help of Reynolds number by calculating loss of energy in the form of pressure drop, friction factor etc.

Devising mixer and reactors

Reynolds number is also significant in mixing. For a mixing tank and reactor Reynolds number is inevitable for efficient mixing and high yield of a reaction.

Scale up or scale down

For a successful scale up or down of a model or prototype, dynamic similarity is required. Reynolds number is critical for dynamic similarity of a model.

Studying Boundary layer

Boundary layer study of a flowing fluid is import in fluid mechanics and Reynolds number helps in that.

Understanding Heat and Mass transport

There many experimental correlations that rely significantly on Reynolds to calculate heat transfer coefficient and mass transfer coefficient.

Q.2 what are the terms used while defining 'Reynolds Number'?

Ans: Terms used in defining Reynolds number are:

System: A system is nothing but a defined boundary under which we do our engineering study. In other words, a small part of any target on which we are going to apply our Mass, Energy, and Momentum transfer fundamentals. For example, a small piece of pipe through which fluid is flowing and fluid behavior is under study

Inertial force: The force we apply on the system or a force applied by surrounding any equipment on the system to cause some movement.

Viscus force: A resisting force by a fluid against the inertial force. Viscus force comes in a fluid due to the entanglement of the particles which resist the motion on the application of any external force.

1. **Viscosity:** Viscosity is a property of a fluid to quantify viscus force (resisting force) that fluid can possess under the action of applied force.

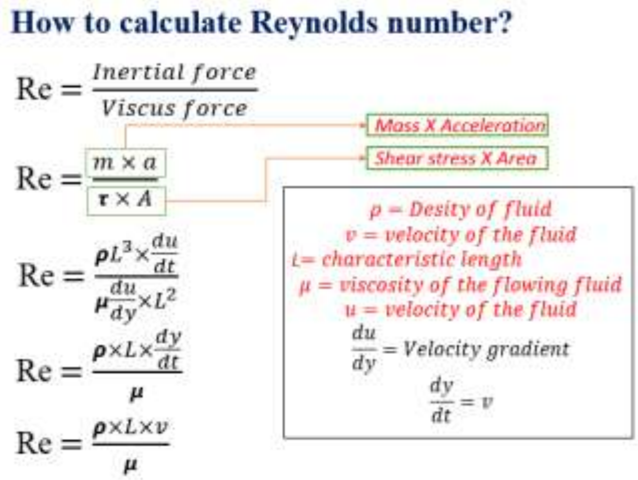
Q.3 Define Reynolds's number?

Ans: Reynolds number is defined as the ratio of inertial force to viscous force. Reynolds number helps us to define the regime in which fluid is flowing-

- Laminar
- Transition
- Turbulent

Q.4 How to calculate Reynolds's number? Show the derivation.

Ans:



Q.5 what are the main features of 'Laminar flow?

Ans:

- Particles follow a stream line path in orderly manner
- Parallel movement of adjacent layers
- Stable flow, No or Negligible intermingling
- First regime to be developed when fluid just starts flowing (relatively at low speeds)
- Less efficient heat and mass transport as compared to other regimes

Q.6 Examples of laminar flow:

Ans:

- Relatively high viscous fluids: Flow of honey or mustard oil from a tilted box
- Water flowing through a small tube at relatively low speed

Q.7 what do you understand by the term 'Flow Regimes'?

Ans:

In fluid mechanics, we deal with fluid at rest in one section and fluid in motion in another part. Whenever fluid is in motion at a certain geometry/system if fluid is moving with fixed velocity and unchanged density and viscosity value, it shows a particular kind of behavior that behavior defines how energy and mass are going to transfer from one specified point to another.

As far as Fluid mechanics is concerned, Study of Fluid flow is divided into 3 main regimes Laminar, Transition and Turbulent based upon behavior of fluid particles.

Fluids' properties such as flow velocity, density, viscosity at a defined characteristic length decides the regime in which its flow will fall. That regime predicts the fate of Heat and Mass transfer.

Q.8 Explain Laminar flow regime?

Ans:

For simplicity, let's say laminar flow has laminate layers that stick over each other. In Laminar flow, fluid particles move in the form of a layer on another layer but don't cross each other and there is no intermixing between the two adjacent layers.

Q.9 Transition flow Regime?

Ans:

It is a regime where fluids possess both Laminar and turbulent flow features. Fluid layers near the wall imparts laminar flow and turbulent at a certain distance away from the wall. In short, Transition regime is a mix of laminar and turbulent flow regime.

Q.10 What are the main features of 'Transition flow?

Ans:

- Fluid follows ill-defined path, chaotic movement
- Unstable flow away from the edges
- In Laminar regime-no mingling, turbulent regime-complete intermingling, but In Transition Regime-Incomplete intermingling
- combination of streamline and random flow directions
- Fluid flow changes between laminar and turbulent flow

Q.11 Examples of Transition flow?

Ans:

- Water flowing in a pipe alternating from smooth to zigzag motion
- Sudden disturbance in orderly flowing fluid

Q.12 Turbulent flow Regime?
Ans:
Turbulent flow is a regime where fluid particles doesn't follow ordered path. Layers of fluid mix with each other randomly and that leads to complete intermingling of layers.

Q.13 Main features of turbulent flow:
Ans:
- Random order of particles in the flowing fluid results formation of eddies, swirls, vortices etc.
- Better intermingling between adjacent layers than other regimes
- Assumed to have complete mixing situation
- Turbulent regime considered to have efficient Heat and mass transport as compared to other regimes
- Turbulent regime generally occurs at relatively high speed and larger systems

Q.14 Examples of turbulent flow:
Ans:
- Relatively high-speed flowing water
- Mixing tanks
- Water flowing through a pipe- internal flow
- Water flowing in a river- open flow

Q.15 Types of Reynolds Number
Ans:
The basic definition of the Reynolds number remains the same that is inertial force to viscous force but where you are applying it varies and leads to 3 main types of Reynolds numbers are-
1. Reynolds number for flowing fluid
2. Reynolds number for particle
3. Reynolds number in mixing

Q.16 what is Reynolds number formula and their units?
Ans:
Reynolds number's formula is Re=(*p*vL)/(dynamic viscosity), where *p=density*, v=velocity, L=Characteristic length. Reynols number has no units because it is a dimensionless number.

Q.17 what is Reynolds number for laminar flow?
Ans:
In a pipe flow, if Reynolds number is <2300 then the fluid flow is considered to be laminar flow.

Q.18 what if Reynolds number is between 2000 and 4000?
Ans:
If Reynolds number is between 2000 and 4000, Fluid flow is in a transition state.

Q.19 how do you know if flow is laminar or turbulent?
Ans:
We can find out whether fluid flow is laminar or turbulent with the help of Reynolds number. For a fluid flowing in a pipe, If Reynolds number value comes out to be <2300 then fluid is flowing in a laminar flow. On the other side, If the Reynolds number value comes out to be >4000 then the fluid is considered to be in a turbulent flow.

Q.20 What are dimensions of Reynolds number?
Ans:
Reynolds is a dimensionless number because it is defined as ration of inertial force to viscus force, dimesions of both forces are same, hence cancelled out for a given system.

Signature :(Joydeep Gupta)
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